THAKUR COLLEGE OF ENGINEERING AND TECHNOLOGY – TCET, MUMBAI

(An Autonomous Institute w.e.f. 2019-20)

A Block, Thakur Educational Campus, Shyamnarayan Thakur Marg, Thakur Village, Kandivali (East), Mumbai – 400101

Affiliation: University of Mumbai | **Approval:** AICTE Approved | **Accreditation:** NAAC A+ Accredited

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[COLLEGE SEAL PLACEHOLDER]

This handbook is a comprehensive guide to the academic policies, programmes, examination schemes, and regulations of Thakur College of Engineering and Technology. This edition includes complete semester-wise course structures for all B.E. and B.Tech. branches offered at the institute. All students are advised to read and understand the contents thoroughly.

1. PROGRAMMES OFFERED

Thakur College of Engineering and Technology offers a wide range of undergraduate and postgraduate programmes across multiple disciplines of engineering, technology, computer application, and management studies.

1.1 Bachelor of Engineering (B.E.) Programmes

S.No.	Programme	Duration	Total Credits	Semesters	Intake
1	B.E. Computer Engineering	4 Years	180	8	As per AICTE/DTE norms
2	B.E. Information Technology	4 Years	180	8	As per AICTE/DTE norms
3	B.E. Electronics & Telecommunication Engg. (EXTC)	4 Years	180	8	As per AICTE/DTE norms
4	B.E. Electronics Engineering	4 Years	180	8	As per AICTE/DTE norms
5	B.E. Mechanical Engineering	4 Years	180	8	As per AICTE/DTE norms
6	B.E. Civil Engineering	4 Years	180	8	As per AICTE/DTE norms
7	B.E. Computer Science & Engineering (CS&E)	4 Years	180	8	As per AICTE/DTE norms
8	B.E. Mechanical & Mechatronics Engg. (MME)	4 Years	180	8	As per AICTE/DTE norms
9	B.E. Electronics & Computer Science (E&CS)	4 Years	180	8	As per AICTE/DTE norms

1.2 Bachelor of Technology (B.Tech.) Programmes

S.No.	Programme	Duration	Total Credits	Semesters	Intake
1	B.Tech. Artificial Intelligence & Data Science (AI&DS)	4 Years	180	8	As per AICTE/DTE norms
2	B.Tech. Artificial Intelligence & Machine Learning (AI&ML)	4 Years	180	8	As per AICTE/DTE norms
3	B.Tech. Internet of Things (IoT)	4 Years	180	8	As per AICTE/DTE norms

1.3 Other Programmes

S.No.	Programme	Duration	Semesters
1	B.Voc. (AI&DS)	3 Years	6
2	B.Voc. (Software Development)	3 Years	6
3	Master of Engineering (M.E.)	2 Years	4

S.No.	Programme	Duration	Semesters
4	Master of Computer Applications (MCA)	2 Years	4
5	Ph.D. (Technology)	--	--

1.4 Eligibility Criteria for B.E./B.Tech. Programmes

Maharashtra State Candidature Candidates:

- Should be a citizen of India.
- Should have passed 10+2 examination or its equivalent with **Physics and Mathematics as compulsory subjects** along with Chemistry or Biotechnology or Biology or Technical Vocational subject or Computer Science or Information Technology or Informatics Practices or Agriculture or Engineering Graphics or Business Studies or Electronics or Entrepreneurship.
- Obtained at least **45% marks** (40% for reserved categories, EWS and PwD candidates belonging to Maharashtra State) in the above subjects taken together.
- Should have appeared in all subjects in **MHT-CET 2024** and obtained non-zero score in PCM MHT-CET.

All India Candidature Candidates:

- Should be an Indian National.
- Should have passed 10+2 with Physics and Mathematics as compulsory subjects along with the above-mentioned subjects and obtained at least 45% marks (40% for reserved categories).
- Obtained **non-zero positive score in JEE (Main)** OR appeared in MHT-CET and obtained non-zero score.

Lateral Entry (Direct Second Year):

- Passed minimum **three years Diploma in Engineering and Technology** with at least 45% marks (40% for reserved categories).
- OR Passed three years Diploma of Vocation (D.Voc.) in the same or allied sector.

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2. ATTENDANCE POLICY

2.1 Minimum Attendance Requirement

A student must maintain a minimum attendance of **75 percent** in the aggregate of all courses taken together in an academic year.

2.2 Condonation of Shortage

The Principal may condone attendance shortage **up to 5 percent** for individual students for valid reasons which must be recorded in writing.

2.3 Detention Rules

Under no circumstances shall a student with **less than 70 percent** aggregate attendance in an academic year be allowed to appear in the semester-end examination.

Students who are detained due to shortage of attendance shall not be promoted to the next academic year. Such students must take re-admission and repeat all courses of the said academic year with the next batch of students.

2.4 Important Provisions

- The Principal shall announce the names of all students who are not eligible to appear in the semester-end examination at least **five calendar days** before the start of the examination.
- Students are expected to be present **five minutes before** the commencement of any scheduled activity including classes, tests, and seminars.
- **Note:** Students are required to complete a minimum **600 hours of internship** from First Year to Final Year.

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3. EXAMINATION AND EVALUATION

3.1 Overview

Thakur College of Engineering and Technology received **Autonomous status** from the Academic Year 2019-20 with six branches of engineering. The institute conducts its own examinations and declares results after moderation by the Board of Examinations.

3.2 Examination Pattern

TCET follows the **Choice Based Credit Grading System with Holistic and Multidisciplinary Education (CBCGS-HME)** .

Component	Marks	Duration
In-Semester Examination (ISE)	20 marks (average of best two out of three ISE tests)	1 hour each
Innovative Examination (IE)	20 marks	---
End Semester Examination (ESE)	60 marks	2 hours
Total	**100 marks**	---

3.3 Marking Scheme

The question paper setting for all examinations is done at the institute level with blind selection of a question paper out of a set of three papers.

3.4 Supplementary Examination

Unsuccessful students are facilitated by introducing a **supplementary examination for theory**, conducted within a month from the declaration of the result, with the same procedure as the End Semester Examination.

3.5 Open House and Revaluation

The **Open House facility** is introduced for transparency, where students can view their answer books and raise grievances, which are resolved. Students are advised to apply for **revaluation** if they are not satisfied

with the assessment in any of the subjects. The revaluation facility has been extended to repeater students from A.Y. 2022-23.

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4. GRADING SYSTEM

4.1 SGPI and CGPI

TCET follows a **10-point grading system** for declaration of results for both undergraduate and postgraduate programmes.

The CGPI is calculated using the formula:

CGPI = Sum of (Grade Points × Credits) / Sum of Credits

4.2 Grade Table

Grade	CGPI Range	Performance Description
O (Outstanding)	10.00	Excellent performance
A+	9.00 – 9.99	Very Good performance
A	8.00 – 8.99	Good performance
B+	7.00 – 7.99	Above Average performance
B	6.00 – 6.99	Average performance
C	5.00 – 5.99	Below Average performance
D	4.00 – 4.99	Marginal performance
F	Below 4.00	Fail

4.3 Pass Percentage

The minimum pass percentage in each paper is **45%** (with a combination of Theory and Sessional marks).

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5. REAPPEAR POLICY (ATKT)

5.1 ATKT Provisions

ATKT (Allowed To Keep Term) is a promotion provision which allows a student to move to the next academic year/semester even if they have a limited number of backlogs (KT).

5.2 ATKT Limits

- **Engineering Programmes:** Students are typically allowed to carry up to **4 ATKT subjects** across semesters for promotion.
- Promotion rules vary based on the specific ordinance for the programme/year.

5.3 Revaluation Process

Students who are not satisfied with the marks awarded may independently apply for **revaluation** to the University through the College in the prescribed form within the period and manner prescribed.

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6. PROMOTION RULES

6.1 First Year to Second Year

Students must clear the required number of subjects as per the ATKT limits prescribed by the University of Mumbai and TCET regulations.

6.2 Second Year to Third Year

Students are promoted with ATKTs up to the allowed limit. If the limit is exceeded, the student must repeat the year. Some programmes require students to clear all First Year ATKTs before entering Third Year.

6.3 Academic Year Break

Students who have not been promoted shall automatically be declared to have taken an Academic Year Break. They must reappear in papers of previous semesters in which they have failed to obtain sufficient credits for promotion.

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7. SUMMER TRAINING AND PROJECT WORK

7.1 Internship Requirements

TCET is the first college in Mumbai to implement a structured internship credit system. All students from First Year to Final Year complete a minimum **600 hours of internship**.

7.2 Summer Internship (First and Second Year)

The summer internship is conducted in the form of **in-house internship** which is mandatory for all students in the summer semester break for **120 hours**. Students may go up to 160 hours to acquire maximum 4 credits.

Activities considered for Summer Internship:

1. Participation in In-house Internship at the end of 1st and 2nd Semester (total 120 hours).
2. Workshop/Training.
3. Festival (Technical/Business/Others) Events.
4. Contribution in Incubation/Innovation/Entrepreneurship Cell/Institutional Innovation Council.
5. Learning at Departmental Lab/Tinkering Lab/Institutional Workshop.

7.3 Internship (Fifth and Sixth Semester)

Students may go up to 240 hours to acquire maximum 6 credits. Students should collectively acquire total contact hours in these activities in a span of 1 year (5th and 6th Semester). A report is submitted to earn Term Work marks in internship at the end of 6th Semester.

Activities considered:

- Participation in in-house internship at the end of 5th and 6th semester (2 weeks each).

- Participation in Hackathon.
- Development of new Product/Business Plan/Registration of start-up.
- Participation in IPR workshop/Leadership talks/Idea/Design/Innovation/Technical Expos.
- Internship with Industry/Govt./NGO/PSU/MSME/Online Internship.
- Long Term Goals under Rural Internship.

7.4 Activity Points

As per AICTE, students have to earn **100 Activity Points** by participating in **400 hours** of activities during 4 years of Engineering. These points will not be reflected in the Grade Card. A separate transcript will be issued to students after completion of Final Year.

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8. AWARD OF DEGREE

8.1 Conditions for Award of Degree

A student shall be eligible for the award of degree subject to the following conditions:

1. The student must have undergone the full course of studies and completed all project reports and training reports specified in the curriculum within the stipulated time.
2. The student must have secured the minimum credits prescribed for the award of the degree (**180 credits** for B.E./B.Tech. programmes).
3. There must be **no outstanding dues** in the student's name to the Institute or University.
4. **No disciplinary action** should be pending against the student.

8.2 Convocation

The annual convocation to confer degrees shall be held in the month of **November** every year.

8.3 Course Completion Certificate

Course completion certificates with information such as medium of instruction of the course and the CGPI obtained by the student are introduced and distributed along with the grade cards.

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9. WORKING HOURS AND ACADEMIC CALENDAR

9.1 Working Hours

The Institute operates from **Monday to Saturday**, from **8:30 AM to 5:30 PM**, excluding Sundays and holidays notified by the Institute.

9.2 Academic Calendar

The Institute follows the academic calendar as laid down by the University of Mumbai. However, the Institute may modify the academic calendar for effective management of the teaching-learning process and for balancing co-curricular and extra-curricular activities. The academic calendar is notified to students through the notice board.

9.3 Examination Schedule

- **In-Semester Examinations (ISE):** Three ISE tests are conducted per semester.
- **End Semester Examinations (ESE):** Conducted at the end of each semester.
- **Supplementary Examinations:** Conducted within a month from the declaration of results.

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10. COURSE STRUCTURE – COMMON FIRST YEAR (SEMESTER I & II)

The First Year curriculum is common for all B.E./B.Tech. programmes at TCET.

Semester I

Code	Paper Name	Credits
BSC-CS101	Engineering Mathematics – I	4
BSC-CS102	Engineering Physics – I	3
BSC-CS103	Engineering Chemistry – I	3
ESC-CS101	Basic Electrical & Electronics Engineering	4
ESC-CS102	Engineering Mechanics	4
ESC-CS103	Engineering Graphics	3
LC-CS101	Physics Lab	2
LC-CS102	Chemistry Lab	2
LC-CS103	Basic Electrical & Electronics Lab	2
Total		**27**

Semester II

Code	Paper Name	Credits
BSC-CS201	Engineering Mathematics – II	4
BSC-CS202	Engineering Physics – II	3
BSC-CS203	Environmental Studies	2
ESC-CS201	Programming for Problem Solving	4
ESC-CS202	Engineering Mechanics – II	3
ESC-CS203	Workshop Practice	3
LC-CS201	Programming Lab	2
LC-CS202	Workshop Lab	2
LC-CS203	Physics Lab – II	2
Total		**25**

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11. COURSE STRUCTURE – B.E. COMPUTER ENGINEERING

Semester III

Code	Paper Name	Credits
BSC-CS301	Mathematics – III	4
PCC-CS301	Data Structures	4
PCC-CS302	Digital Logic Design	3
PCC-CS303	Computer Organization & Architecture	3
PCC-CS304	Object Oriented Programming	3
LC-CS301	Data Structures Lab	2
LC-CS302	Digital Logic Design Lab	2
LC-CS303	OOP Lab	2
Total		**23**

Semester IV

Code	Paper Name	Credits
BSC-CS401	Mathematics – IV	4
PCC-CS401	Database Management Systems	4
PCC-CS402	Operating Systems	4
PCC-CS403	Analysis of Algorithms	3
PCC-CS404	Software Engineering	3
LC-CS401	DBMS Lab	2
LC-CS402	OS Lab	2
LC-CS403	Algorithms Lab	2
Total		**24**

Semester V

Code	Paper Name	Credits
PCC-CS501	Computer Networks	4
PCC-CS502	Theory of Computation	3
PCC-CS503	Microprocessors	3
PEC-CS501	Professional Elective – I	3

Code	Paper Name	Credits
OEC-CS501	Open Elective – I	3
LC-CS501	Computer Networks Lab	2
LC-CS502	Microprocessors Lab	2
PROJ-CS501	Mini Project	2
Total		**22**

Professional Elective – I Options:

- Advanced Data Structures
- Web Technologies
- Java Programming

Semester VI

Code	Paper Name	Credits
PCC-CS601	Compiler Construction	3
PCC-CS602	Cryptography & Network Security	3
PEC-CS601	Professional Elective – II	3
PEC-CS602	Professional Elective – III	3
OEC-CS601	Open Elective – II	3
LC-CS601	Compiler Lab	2
PROJ-CS601	Major Project – Phase I	3
Total		**20**

Professional Elective – II Options:

- Internet Programming
- Data Warehousing & Mining
- Digital Signal Processing
- Soft Computing

Professional Elective – III Options:

- Software Process Automation
- Entrepreneurship Development and Management
- Cyber Security and Laws
- Reliability Engineering

Semester VII

Code	Paper Name	Credits
PCC-CS701	Cloud Computing	3
PCC-CS702	Artificial Intelligence	3
PEC-CS701	Professional Elective – IV	3
PEC-CS702	Professional Elective – V	3
OEC-CS701	Open Elective – III	3
LC-CS701	AI Lab	2
PROJ-CS701	Major Project – Phase II	4
Total		**21**

Professional Elective – IV Options:

- Machine Learning
- Advanced Computer Networks
- Image Processing

Professional Elective – V Options:

- Natural Language Processing
- Blockchain Technology
- Big Data Analytics

Semester VIII

Code	Paper Name	Credits
PROJ-CS801	Major Project – Phase III	6
LC-CS801	Seminar	2
Total		**8**

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12. COURSE STRUCTURE – B.E. INFORMATION TECHNOLOGY

Semester III

Code	Paper Name	Credits
BSC-IT301	Mathematics – III	4
PCC-IT301	Data Structures and Analysis	4
PCC-IT302	Digital Electronics	4
PCC-IT303	Computer Organization & Architecture	4
PCC-IT304	Object Oriented Programming	3

Code	Paper Name	Credits
LC-IT301	Data Structures Lab	2
LC-IT302	Digital Electronics Lab	2
LC-IT303	OOP Lab	2
Total		**25**

Semester IV

Code	Paper Name	Credits
BSC-IT401	Mathematics – IV	4
PCC-IT401	Database Management Systems	4
PCC-IT402	Operating Systems	4
PCC-IT403	Principles of Communication	4
PCC-IT404	Software Engineering	3
LC-IT401	DBMS Lab	2
LC-IT402	OS Lab	2
LC-IT403	Communication Lab	2
Total		**25**

Semester V

Code	Paper Name	Credits
PCC-IT501	Computer Networks	4
PCC-IT502	Web Technologies	4
PCC-IT503	Microprocessors	3
PEC-IT501	Professional Elective – I	3
OEC-IT501	Open Elective – I	3
LC-IT501	Computer Networks Lab	2
LC-IT502	Web Technologies Lab	2
PROJ-IT501	Mini Project	2
Total		**23**

Semester VI

Code	Paper Name	Credits
PCC-IT601	Cloud Computing	3
PCC-IT602	Cybersecurity	3
PEC-IT601	Professional Elective – II	3
PEC-IT602	Professional Elective – III	3
OEC-IT601	Open Elective – II	3
LC-IT601	Cybersecurity Lab	2
PROJ-IT601	Major Project – Phase I	3
Total		**20**

Semester VII

Code	Paper Name	Credits
PCC-IT701	Data Analytics	3
PCC-IT702	Artificial Intelligence & ML	3
PEC-IT701	Professional Elective – IV	3
PEC-IT702	Professional Elective – V	3
OEC-IT701	Open Elective – III	3
LC-IT701	AI & ML Lab	2
PROJ-IT701	Major Project – Phase II	4
Total		**21**

Semester VIII

Code	Paper Name	Credits
PROJ-IT801	Major Project – Phase III	6
LC-IT801	Seminar	2
Total		**8**

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13. COURSE STRUCTURE – B.E. ELECTRONICS & TELECOMMUNICATION (EXTC)

Semester III

Code	Paper Name	Credits
BSC-EXTC301	Mathematics – III	4

Code	Paper Name	Credits
PCC-EXTC301	Electronic Devices & Circuits	4
PCC-EXTC302	Digital Electronics	4
PCC-EXTC303	Network Theory	3
PCC-EXTC304	Signals & Systems	3
LC-EXTC301	Electronic Devices Lab	2
LC-EXTC302	Digital Electronics Lab	2
LC-EXTC303	Network Theory Lab	2
Total		**24**

Semester IV

Code	Paper Name	Credits
BSC-EXTC401	Mathematics – IV	4
PCC-EXTC401	Analog Circuits	4
PCC-EXTC402	Communication Systems	4
PCC-EXTC403	Microprocessors & Microcontrollers	4
PCC-EXTC404	Electromagnetic Waves	3
LC-EXTC401	Analog Circuits Lab	2
LC-EXTC402	Communication Systems Lab	2
LC-EXTC403	Microprocessors Lab	2
Total		**25**

Semester V

Code	Paper Name	Credits
PCC-EXTC501	Digital Signal Processing	4
PCC-EXTC502	Computer Networks	4
PCC-EXTC503	Control Systems	3
PEC-EXTC501	Professional Elective – I	3
OEC-EXTC501	Open Elective – I	3
LC-EXTC501	DSP Lab	2
LC-EXTC502	Computer Networks Lab	2

Code	Paper Name	Credits
PROJ-EXTC501	Mini Project	2
Total		**23**

Semester VI

Code	Paper Name	Credits
PCC-EXTC601	Wireless Communication	4
PCC-EXTC602	VLSI Design	3
PEC-EXTC601	Professional Elective – II	3
PEC-EXTC602	Professional Elective – III	3
OEC-EXTC601	Open Elective – II	3
LC-EXTC601	VLSI Lab	2
PROJ-EXTC601	Major Project – Phase I	3
Total		**21**

Semester VII

Code	Paper Name	Credits
PCC-EXTC701	Microwave Engineering	3
PCC-EXTC702	Optical Communication	3
PEC-EXTC701	Professional Elective – IV	3
PEC-EXTC702	Professional Elective – V	3
OEC-EXTC701	Open Elective – III	3
LC-EXTC701	Microwave Lab	2
PROJ-EXTC701	Major Project – Phase II	4
Total		**21**

Semester VIII

Code	Paper Name	Credits
PROJ-EXTC801	Major Project – Phase III	6
LC-EXTC801	Seminar	2
Total		**8**

14. COURSE STRUCTURE – B.E. MECHANICAL ENGINEERING

Semester III

Code	Paper Name	Credits
BSC-MECH301	Mathematics – III	4
PCC-MECH301	Thermodynamics	4
PCC-MECH302	Strength of Materials	4
PCC-MECH303	Fluid Mechanics	4
PCC-MECH304	Materials Science	3
LC-MECH301	Thermodynamics Lab	2
LC-MECH302	Strength of Materials Lab	2
LC-MECH303	Fluid Mechanics Lab	2
Total		**25**

Semester IV

Code	Paper Name	Credits
BSC-MECH401	Mathematics – IV	4
PCC-MECH401	Manufacturing Processes	4
PCC-MECH402	Theory of Machines	4
PCC-MECH403	Mechanical Vibrations	3
PCC-MECH404	Engineering Drawing	3
LC-MECH401	Manufacturing Lab	2
LC-MECH402	Theory of Machines Lab	2
LC-MECH403	Mechanical Vibrations Lab	2
Total		**24**

Semester V

Code	Paper Name	Credits
PCC-MECH501	Machine Design – I	4
PCC-MECH502	Dynamics of Machines	4
PCC-MECH503	Heat & Mass Transfer	4
PEC-MECH501	Professional Elective – I	3

Code	Paper Name	Credits
OEC-MECH501	Open Elective – I	3
LC-MECH501	Machine Design Lab	2
LC-MECH502	Heat Transfer Lab	2
PROJ-MECH501	Mini Project	2
Total		**24**

Semester VI

Code	Paper Name	Credits
PCC-MECH601	Machine Design – II	3
PCC-MECH602	Production Process	3
PEC-MECH601	Professional Elective – II	3
PEC-MECH602	Professional Elective – III	3
OEC-MECH601	Open Elective – II	3
LC-MECH601	Production Lab	2
PROJ-MECH601	Major Project – Phase I	3
Total		**20**

Semester VII

Code	Paper Name	Credits
PCC-MECH701	Material Technology	3
PCC-MECH702	Manufacturing Planning and Control	3
PEC-MECH701	Professional Elective – IV	3
PEC-MECH702	Professional Elective – V	3
OEC-MECH701	Open Elective – III	3
LC-MECH701	Material Technology Lab	2
PROJ-MECH701	Major Project – Phase II	4
Total		**21**

Semester VIII

Code	Paper Name	Credits
PROJ-MECH801	Major Project – Phase III	6

Code	Paper Name	Credits
LC-MECH801	Seminar	2
Total		**8**

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15. COURSE STRUCTURE – B.E. CIVIL ENGINEERING

Semester III

Code	Paper Name	Credits
BSC-CIV301	Mathematics – III	4
PCC-CIV301	Mechanics of Structures – I	4
PCC-CIV302	Fluid Mechanics – I	4
PCC-CIV303	Building Materials	4
PCC-CIV304	Surveying – I	3
LC-CIV301	Fluid Mechanics Lab	2
LC-CIV302	Surveying Lab – I	2
LC-CIV303	Building Materials Lab	2
Total		**25**

Semester IV

Code	Paper Name	Credits
BSC-CIV401	Mathematics – IV	4
PCC-CIV401	Mechanics of Structures – II	4
PCC-CIV402	Geotechnical Engineering – I	4
PCC-CIV403	Surveying – II	4
PCC-CIV404	Structural Analysis – I	3
LC-CIV401	Geotechnical Lab	2
LC-CIV402	Surveying Lab – II	2
LC-CIV403	Structural Analysis Lab	2
Total		**25**

Semester V

Code	Paper Name	Credits
PCC-CIV501	Structural Engineering – I	4
PCC-CIV502	Transportation Engineering – I	4
PCC-CIV503	Environmental Engineering – I	4
PEC-CIV501	Professional Elective – I	3
OEC-CIV501	Open Elective – I	3
LC-CIV501	Structural Engineering Lab	2
LC-CIV502	Transportation Lab	2
PROJ-CIV501	Mini Project	2
Total		**24**

Semester VI

Code	Paper Name	Credits
PCC-CIV601	Structural Engineering – II	3
PCC-CIV602	Construction Technology	3
PEC-CIV601	Professional Elective – II	3
PEC-CIV602	Professional Elective – III	3
OEC-CIV601	Open Elective – II	3
LC-CIV601	Construction Lab	2
PROJ-CIV601	Major Project – Phase I	3
Total		**20**

Semester VII

Code	Paper Name	Credits
PCC-CIV701	Estimating and Costing	3
PCC-CIV702	Design of Steel Structures	3
PEC-CIV701	Professional Elective – IV	3
PEC-CIV702	Professional Elective – V	3
OEC-CIV701	Open Elective – III	3
LC-CIV701	Steel Structures Lab	2
PROJ-CIV701	Major Project – Phase II	4

Code	Paper Name	Credits
Total		**21**

Semester VIII

Code	Paper Name	Credits
PROJ-CIV801	Major Project – Phase III	6
LC-CIV801	Seminar	2
Total		**8**

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16. COURSE STRUCTURE – B.TECH. ARTIFICIAL INTELLIGENCE & DATA SCIENCE (AI&DS)

Semester III

Code	Paper Name	Credits
BSC-AIDS301	Mathematics – III	4
PCC-AIDS301	Data Structures & Algorithms	4
PCC-AIDS302	Database Management Systems	3
PCC-AIDS303	Object Oriented Programming	3
PCC-AIDS304	Digital Logic & Computer Organization	3
LC-AIDS301	Data Structures Lab	2
LC-AIDS302	DBMS Lab	2
LC-AIDS303	OOP Lab	2
Total		**23**

Semester IV

Code	Paper Name	Credits
BSC-AIDS401	Mathematics – IV (Linear Algebra)	4
PCC-AIDS401	Introduction to Artificial Intelligence	4
PCC-AIDS402	Python Programming	3
PCC-AIDS403	Operating Systems	3
PCC-AIDS404	Software Engineering	3
LC-AIDS401	AI Lab	2
LC-AIDS402	Python Lab	2

Code	Paper Name	Credits
LC-AIDS403	OS Lab	2
Total		**23**

Semester V

Code	Paper Name	Credits
PCC-AIDS501	Machine Learning	4
PCC-AIDS502	Data Science	4
PCC-AIDS503	Computer Networks	3
PEC-AIDS501	Professional Elective – I	3
OEC-AIDS501	Open Elective – I	3
LC-AIDS501	Machine Learning Lab	2
LC-AIDS502	Data Science Lab	2
PROJ-AIDS501	Mini Project	2
Total		**23**

Semester VI

Code	Paper Name	Credits
PCC-AIDS601	Deep Learning	4
PCC-AIDS602	Big Data Analytics	3
PEC-AIDS601	Professional Elective – II	3
PEC-AIDS602	Professional Elective – III	3
OEC-AIDS601	Open Elective – II	3
LC-AIDS601	Deep Learning Lab	2
PROJ-AIDS601	Major Project – Phase I	3
Total		**21**

Semester VII

Code	Paper Name	Credits
PCC-AIDS701	Natural Language Processing	3
PCC-AIDS702	Computer Vision	3
PEC-AIDS701	Professional Elective – IV	3

Code	Paper Name	Credits
PEC-AIDS702	Professional Elective – V	3
OEC-AIDS701	Open Elective – III	3
LC-AIDS701	NLP & CV Lab	2
PROJ-AIDS701	Major Project – Phase II	4
Total		**21**

Semester VIII

Code	Paper Name	Credits
PROJ-AIDS801	Major Project – Phase III	6
LC-AIDS801	Seminar	2
Total		**8**

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17. COURSE STRUCTURE – B.TECH. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (AI&ML)

Semester III

Code	Paper Name	Credits
BSC-AIML301	Mathematics – III	4
PCC-AIML301	Data Structures & Algorithms	4
PCC-AIML302	Database Management Systems	3
PCC-AIML303	Object Oriented Programming	3
PCC-AIML304	Digital Logic & Computer Organization	3
LC-AIML301	Data Structures Lab	2
LC-AIML302	DBMS Lab	2
LC-AIML303	OOP Lab	2
Total		**23**

Semester IV

Code	Paper Name	Credits
BSC-AIML401	Mathematics – IV (Linear Algebra & Probability)	4
PCC-AIML401	Introduction to AI & ML	4
PCC-AIML402	Python Programming	3

Code	Paper Name	Credits
PCC-AIML403	Operating Systems	3
PCC-AIML404	Software Engineering	3
LC-AIML401	AI & ML Lab	2
LC-AIML402	Python Lab	2
LC-AIML403	OS Lab	2
Total		**23**

Semester V

Code	Paper Name	Credits
PCC-AIML501	Machine Learning	4
PCC-AIML502	Data Science	3
PCC-AIML503	Computer Networks	3
PEC-AIML501	Professional Elective – I	3
OEC-AIML501	Open Elective – I	3
LC-AIML501	Machine Learning Lab	2
LC-AIML502	Data Science Lab	2
PROJ-AIML501	Mini Project	2
Total		**22**

Semester VI

Code	Paper Name	Credits
PCC-AIML601	Deep Learning	4
PCC-AIML602	Reinforcement Learning	3
PEC-AIML601	Professional Elective – II	3
PEC-AIML602	Professional Elective – III	3
OEC-AIML601	Open Elective – II	3
LC-AIML601	Deep Learning Lab	2
PROJ-AIML601	Major Project – Phase I	3
Total		**21**

Semester VII

Code	Paper Name	Credits
PCC-AIML701	Natural Language Processing	3
PCC-AIML702	Computer Vision	3
PEC-AIML701	Professional Elective – IV	3
PEC-AIML702	Professional Elective – V	3
OEC-AIML701	Open Elective – III	3
LC-AIML701	NLP & CV Lab	2
PROJ-AIML701	Major Project – Phase II	4
Total		**21**

Semester VIII

Code	Paper Name	Credits
PROJ-AIML801	Major Project – Phase III	6
LC-AIML801	Seminar	2
Total		**8**

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18. ADDITIONAL IMPORTANT INFORMATION

18.1 Open Elective Courses (Offered Across Branches)

S.No.	Open Elective Code	Course Name
1	OEC-01	Entrepreneurship Development
2	OEC-02	Financial Management
3	OEC-03	Environmental Science & Sustainability
4	OEC-04	Cyber Security & Laws
5	OEC-05	Human Values & Ethics
6	OEC-06	Industrial Psychology
7	OEC-07	Project Management
8	OEC-08	Business Analytics

18.2 Mandatory Non-Credit Courses (NUES)

S.No.	Course Name	Semester
1	Environmental Studies	Semester II

S.No.	Course Name	Semester
2	Cyber Ethics	Semester IV
3	Personality Development Skills	Semester VI
4	Seminar	Semester VIII

18.3 Internship Credit Distribution

Year	Internship Hours	Credits
First Year (Summer)	120 hours	4
Second Year (Summer)	120 hours	4
Third Year (5th & 6th Semester)	240 hours	6
Fourth Year (7th & 8th Semester)	120 hours	4
Total	**600 hours**	**18**

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[COLLEGE SEAL PLACEHOLDER]

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